

Technical Report: Diagnostic Analysis using Python

1. Context

Between January 2020 and June 2022, the NHS in England faced major operational pressures due to the COVID-19 pandemic, leading to unpredictable changes in patient demand and behaviour.

To allocate resources effectively and plan for the future, the NHS requires clear, evidence-based insights into whether staff and infrastructure are sufficient, and how the service capacity is actually utilised.

This analysis aims to reveal trends in NHS service capacity utilisation and missed appointments (DNA), thereby supporting informed decision-making and improved patient care during a period of significant operational challenges.

Business Objective: To optimise NHS service efficiency by analysing missed appointments (DNA) and service capacity utilisation trends.

2. Analytical Approach

This analysis was conducted using Python within a Jupyter Notebook, following a systematic workflow to import, clean, and analyse NHS service data. The aim was to prepare reliable datasets suitable for identifying trends in service capacity utilisation and missed appointments (DNA).

1. Data Import

The three provided datasets were imported using the Pandas library. Each file was read into a DataFrame, and initial inspections were conducted to verify column names, data types, and completeness. Date columns were standardised by converting them to a datetime format to ensure consistency in time-based analyses.

```
Prepare your workstation

[1]: # Import the necessary libraries.
import pandas as pd
import numpy as np

# Optional - Ignore warnings.
import warnings
warnings.filterwarnings('ignore')

[2]: # Import and sense-check the 'actual_duration.csv' file.
ad = pd.read_csv('actual_duration.csv')
# View the DataFrame.
ad
```

2. Data Cleaning and Preparation

A thorough quality assessment was completed, involving the detection and quantification of missing values, removal of duplicate records and conversion of data types of variables suitable for analysis.

Merges between DataFrames were not made to avoid duplicates, since the appointments were records with no unique ID numbers.

The majority of the data analysis was done using the 'appointments_regional' DataFrame, as it covered data over 30 months. (January 2020 – June 2022)

The 'appointment_date' and 'appointment_month' attributes were converted to datetime format for analysis.

Data across all three DataFrames were chosen to answer specific questions to support successful business outcomes, while excluding unknowns. DataFrames were checked and sense-checked before producing visualisations.

```
[8]: # Determine whether there are missing values.
ar_missing = ar[ar.isna().values.any(axis = 1)]

# View ar_missing
ar_missing

[8]: icb_ons_code  appointment_month  appointment_status  hcp_type  appointment_mode  time_between_book_and_appointment  count_of_appointments

21604 duplicates were found in the 'appointments_regional.csv' file.

[9]: # Check for duplicate records in appointments_regional.
duplicates_ar = ar[ar.duplicated()].sum()
value_counts = ar.duplicated(keep=False).value_counts()
print("Breakdown of unique vs duplicate rows:")
print(value_counts)

# Remove 21604 duplicates from appointments_regional and keep the unique rows.
ar_unique = ar.drop_duplicates()
print(f"Original DataFrame shape: {ar.shape}")
print(f"DataFrame after removing duplicates: {ar_unique.shape}")

Breakdown of unique vs duplicate rows:
False    556686
True      40135
Name: count, dtype: int64
Original DataFrame shape: (596821, 7)
DataFrame after removing duplicates: (575217, 7)
```

```
[7]: # Import and sense-check the appointments_regional.csv data set as ar.
ar = pd.read_csv('appointments_regional.csv')

# View the DataFrame.
ar

[7]:
```

	icb_ons_code	appointment_month	appointment_status	hcp_type	appointment_mode	time_between_book_and_appointment	count_of_appointments
0	E54000034	2020-01	Attended	GP	Face-to-Face	1 Day	8107
1	E54000034	2020-01	Attended	GP	Face-to-Face	15 to 21 Days	6791
2	E54000034	2020-01	Attended	GP	Face-to-Face	2 to 7 Days	20686
3	E54000034	2020-01	Attended	GP	Face-to-Face	22 to 28 Days	4268
4	E54000034	2020-01	Attended	GP	Face-to-Face	8 to 14 Days	11971
...	...	...	...	...	...	...	...
596816	E54000050	2022-06	Unknown	Unknown	Unknown	2 to 7 Days	21
596817	E54000050	2022-06	Unknown	Unknown	Unknown	22 to 28 Days	8
596818	E54000050	2022-06	Unknown	Unknown	Unknown	8 to 14 Days	28
596819	E54000050	2022-06	Unknown	Unknown	Unknown	More than 28 Days	17
596820	E54000050	2022-06	Unknown	Unknown	Unknown	Same Day	10

596821 rows x 7 columns

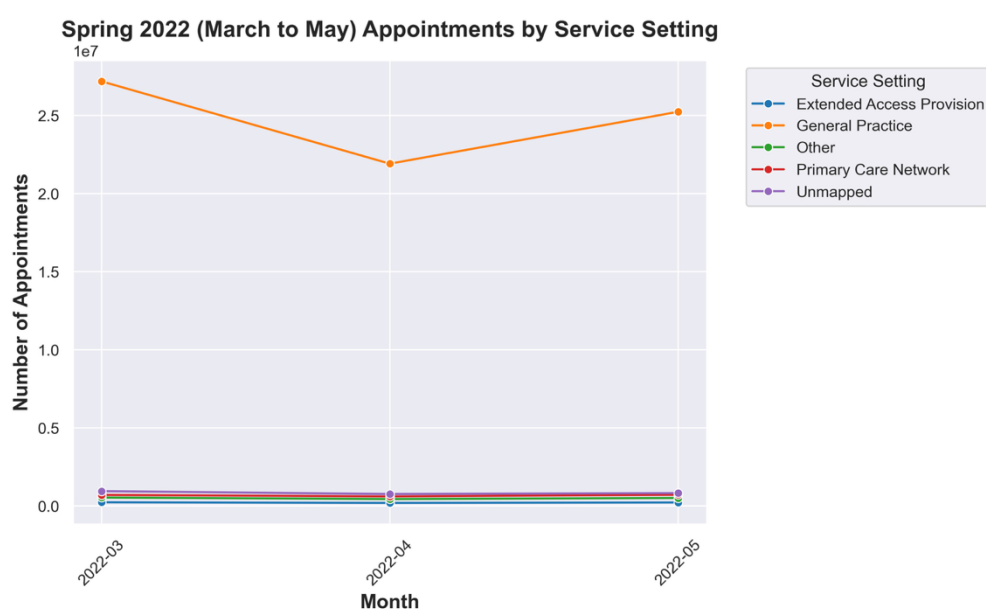
No missing values were found in ar.

3. Visualisation and Insights

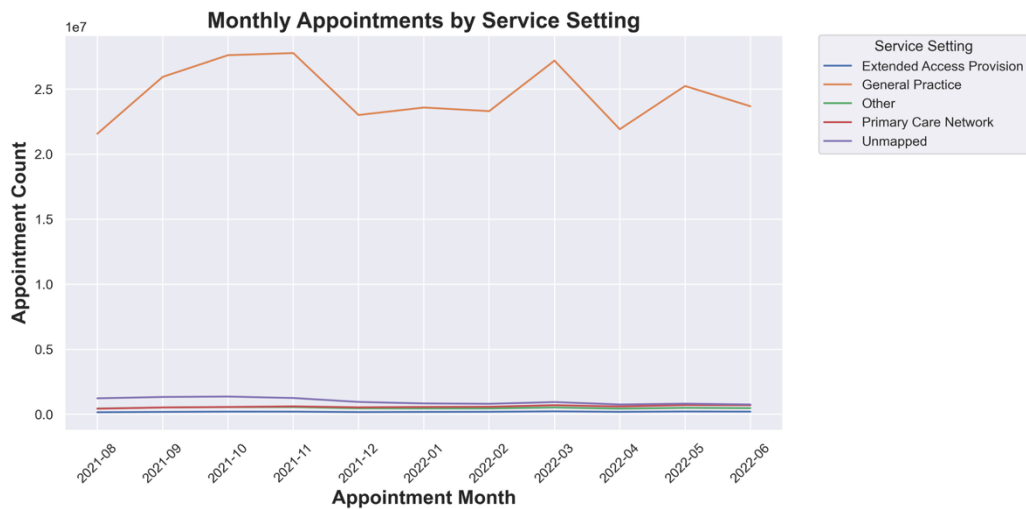
3.1 Monthly and Seasonal Appointment Trends



The line chart helped assess the monthly appointment trends over the 30 months (January 2020 – June 2022) and uncover the effects of external factors such as the COVID-19 pandemic waves and seasonal trends (Christmas, New Year and Easter holidays). The number of appointments ranged from 27 million to over 30 million per month. The month of November 2021 had the highest volume of appointments in total (30395923).



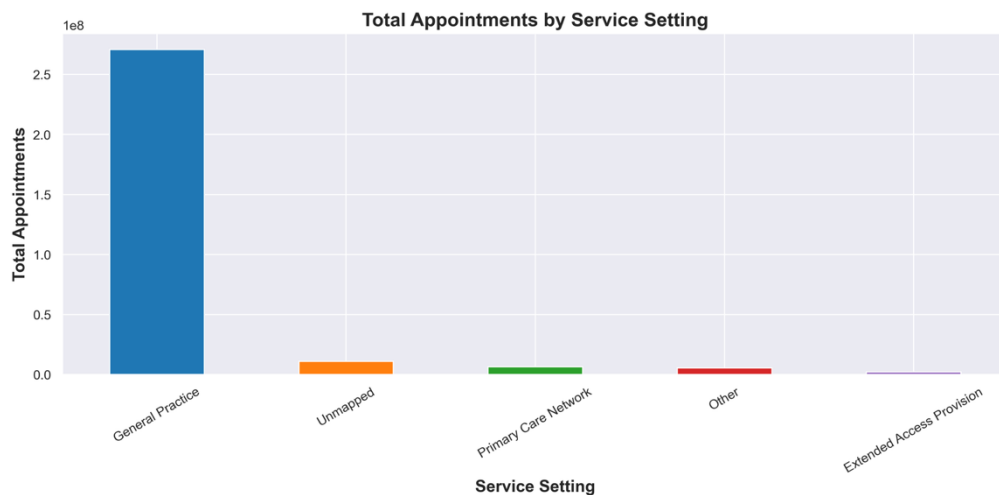
The Spring 2022 chart showcased a very noticeable dip in the number of appointments pre-Easter holidays, and a gradual increase post-Easter holidays. (Appendix 3.3.)

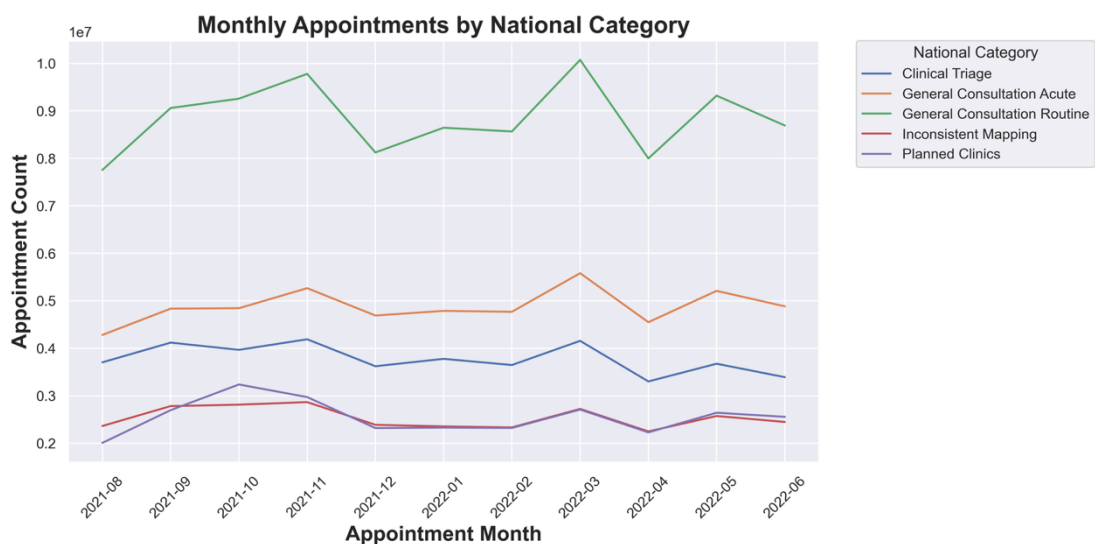
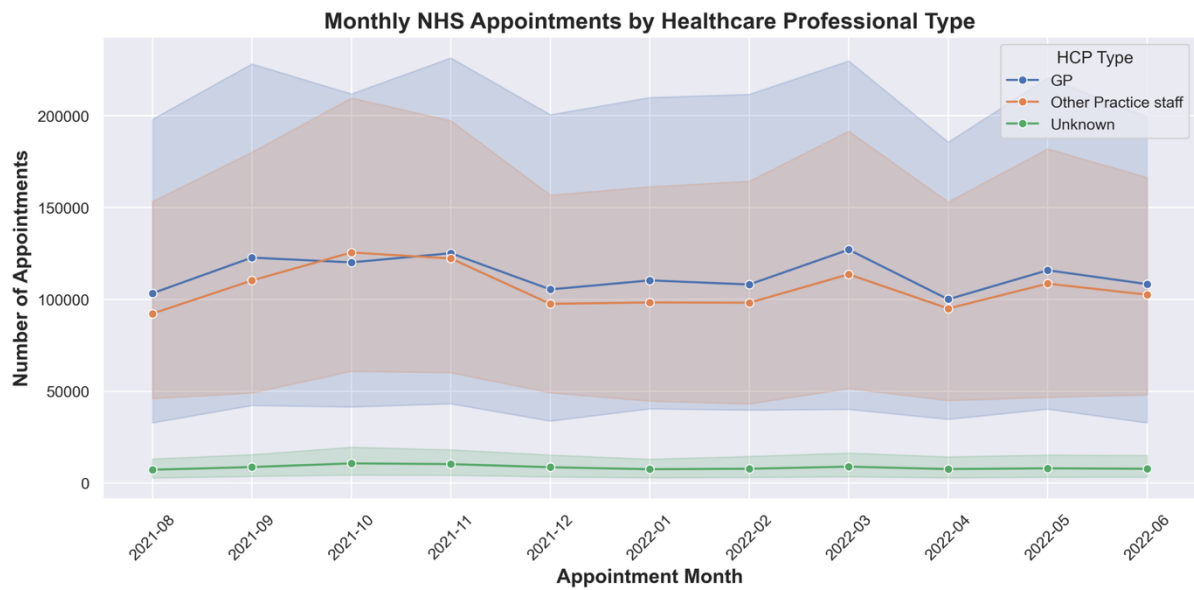


Insight: Seasonal trends in appointments suggest that the NHS should prepare well in advance for the high volume of appointments and allocate the staffing and resources accordingly. In addition to seasonal trends, the appointments were influenced by the COVID-19 pandemic waves: the first wave during March-April 2020, the second wave during September-November 2020 and the third during January-February 2021.

2.2 Service Setting, Context Type and National Category Statistics

General Practices were the most popular service settings for appointments, regardless of the season or location, catering to about 91.5% of the total appointments.





Insight: Since General Practices are the most popular for appointments, the NHS could focus on reallocating resources and sufficiently staffing these settings. The majority of the appointment activity is concentrated in Care Related Encounters; it is critical that staffing, funding, and operational planning are closely aligned to support these areas. This would allow for ease of management when appointments are high in volume.

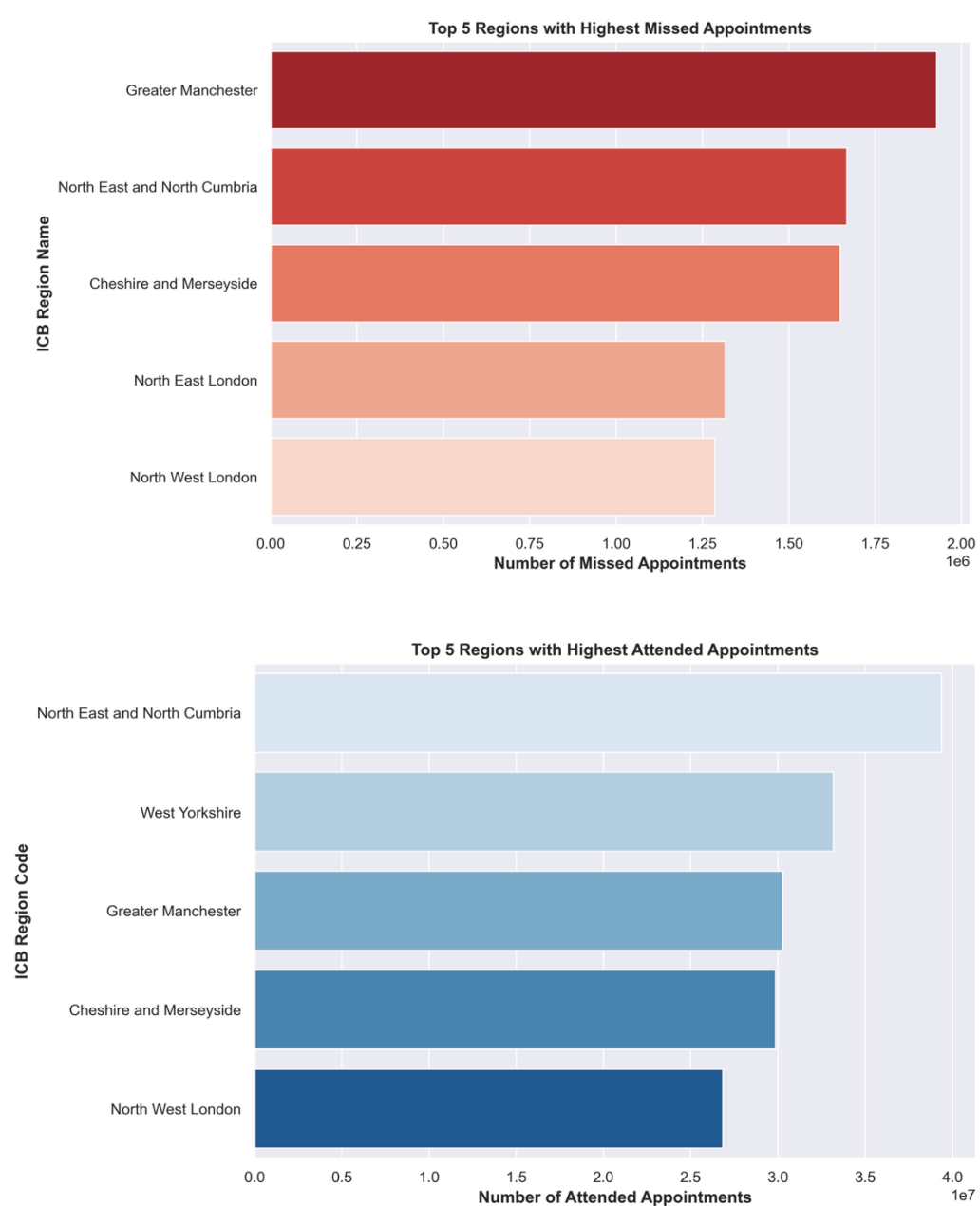
Additionally, improving the data quality by reducing the number of unmapped and inconsistently mapped appointments would help uncover valuable insights and depict the granularity of the data.

2.3 Location Statistics

Some ICB locations, like North East and North Cumbria, had extremely low missed appointments (DNA) (DNA), while locations like Greater Manchester had higher ‘DNA’ rates.

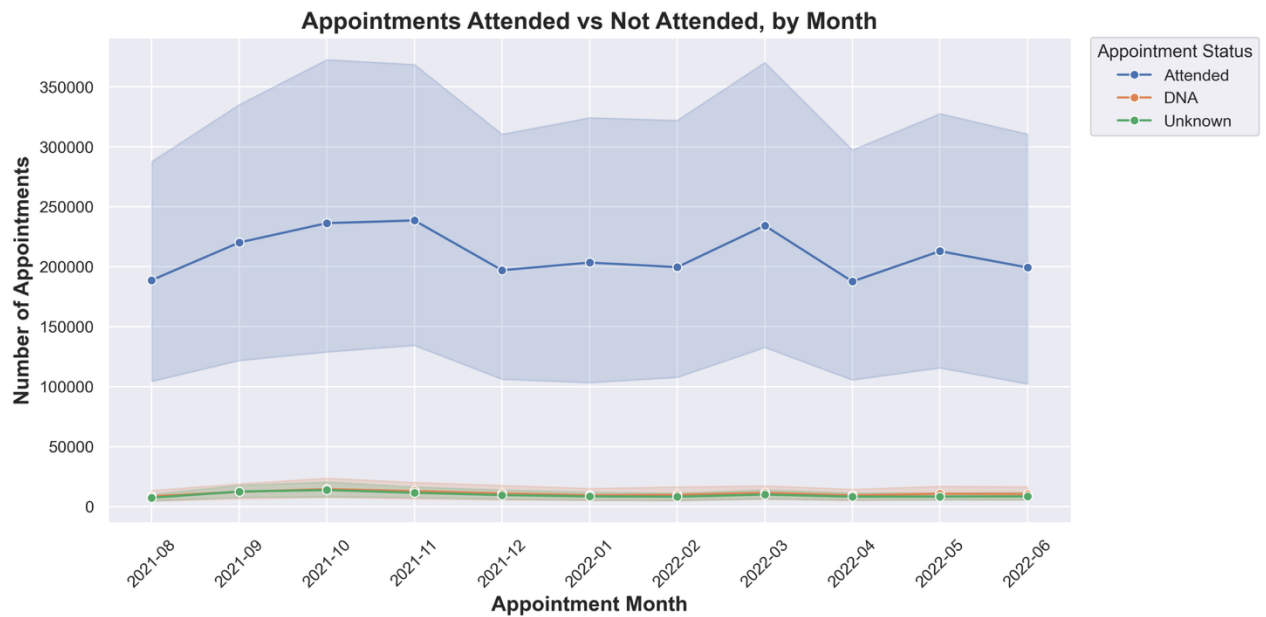
Insight: Identifying patient behaviour in some areas, such as Greater Manchester, would give rise to underlying factors such as socioeconomic barriers, communication gaps, access or booking issues.

Promoting a hybrid-care model in such areas could provide access to remote healthcare. Additionally, studying models adopted by high-performing locations like North East and North Cumbria would also lead to reduced missed appointments (DNA).

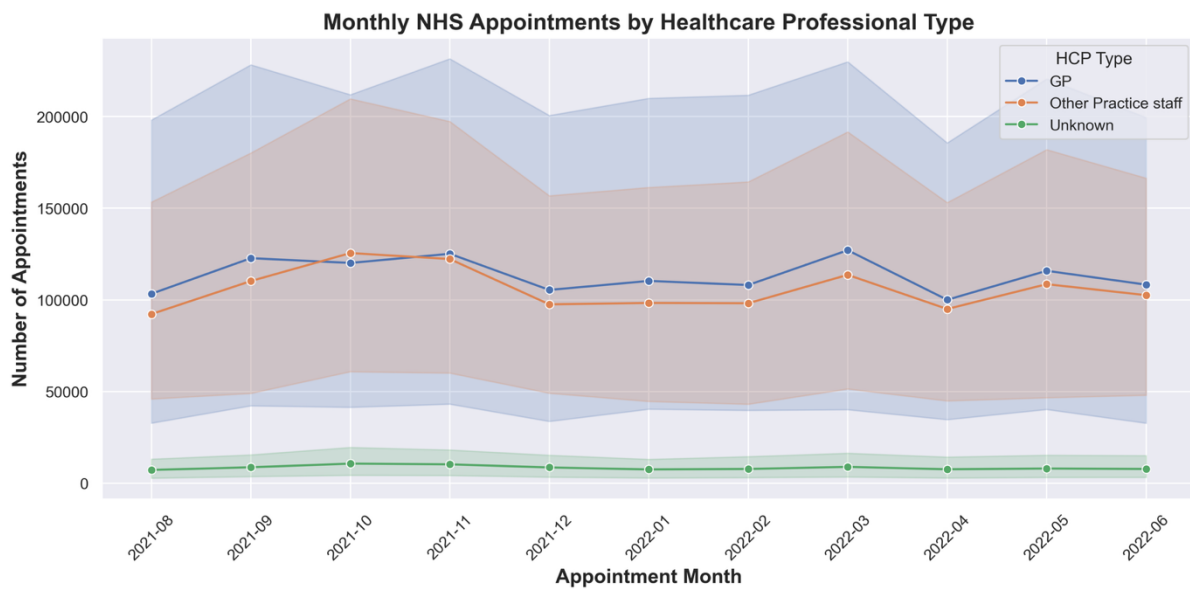


3.4 Appointment Statistics

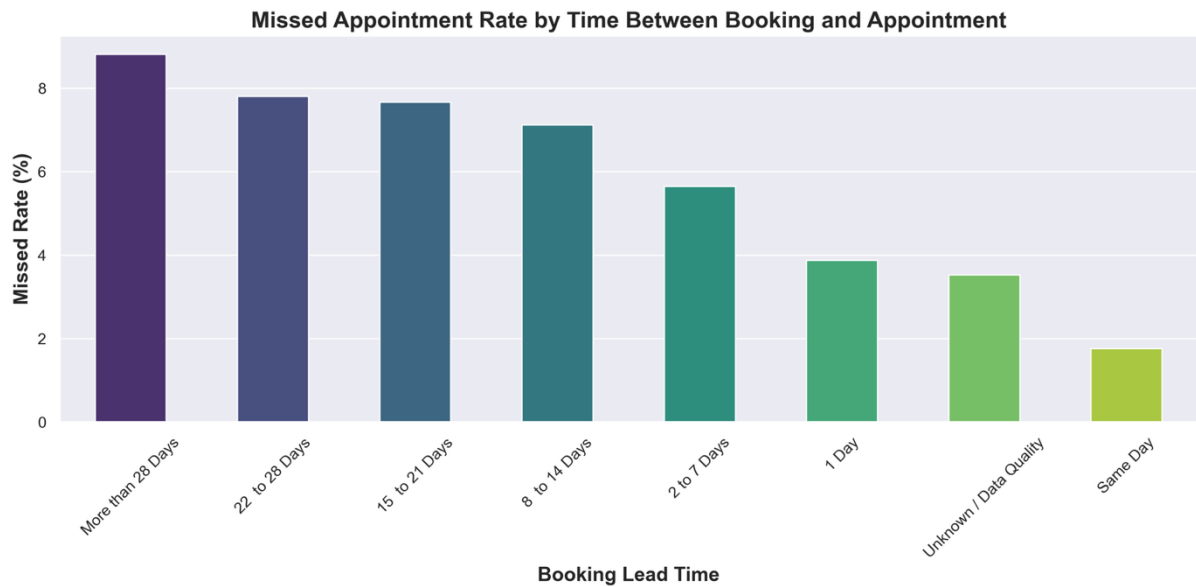
- About 4.4% of appointments were missed overall.



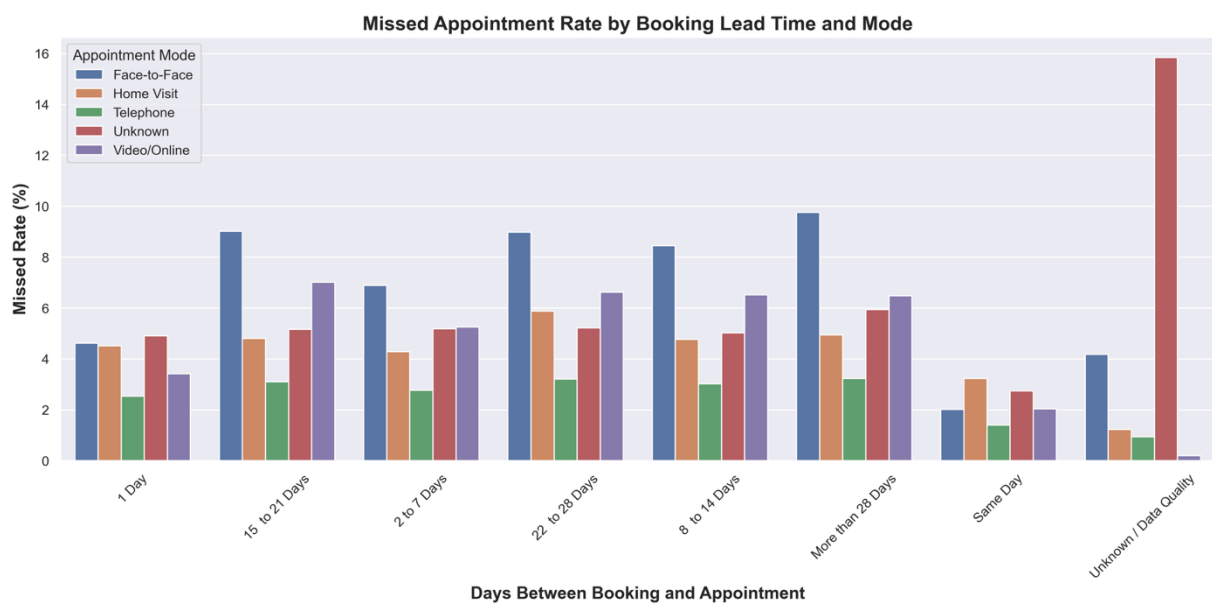
- GPs seemed to be the most preferred HealthCare professionals in NHS appointments.



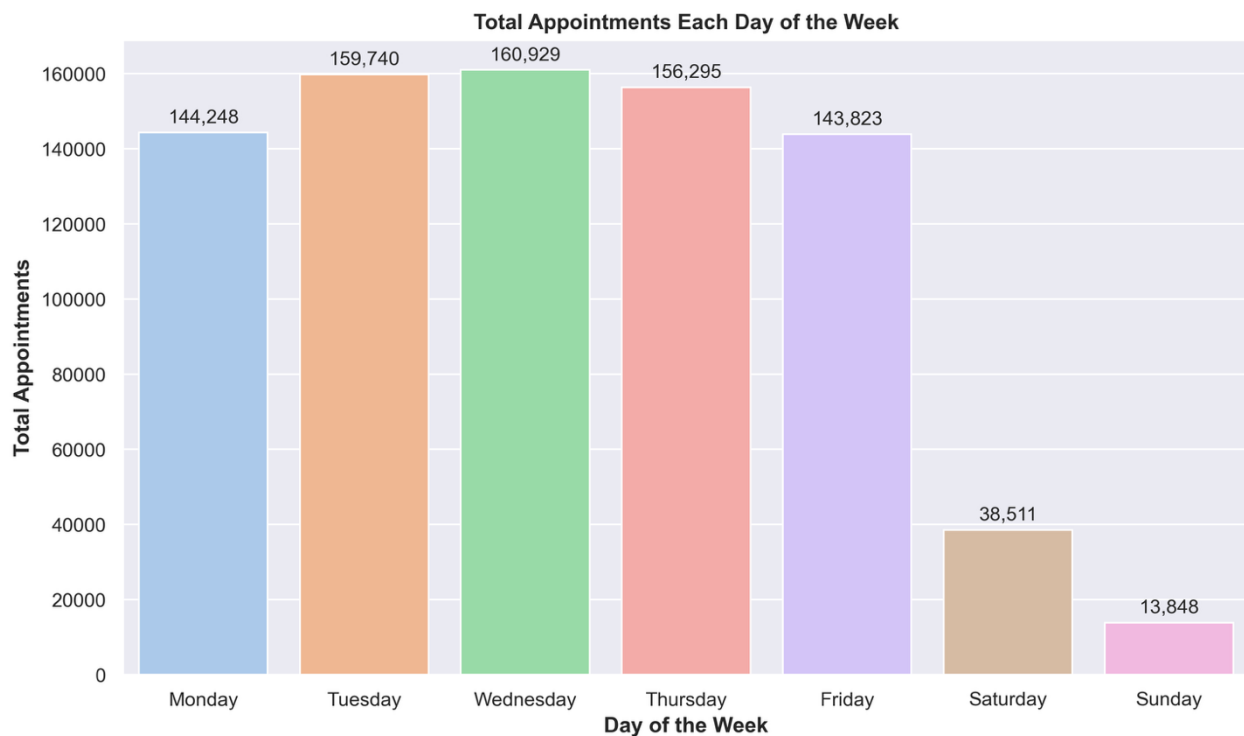
- Appointment bookings made the same day as the appointment had the lowest missed appointment rates, as shown in the bar plot.



- 'Face-to-Face' appointment modes had comparatively higher missed (DNA) rates.



- Appointments were higher in number on weekdays and lower during the weekends. The bar plot shows the number of appointments on each day of the week, while the line plot shows the total weekly appointment trend.

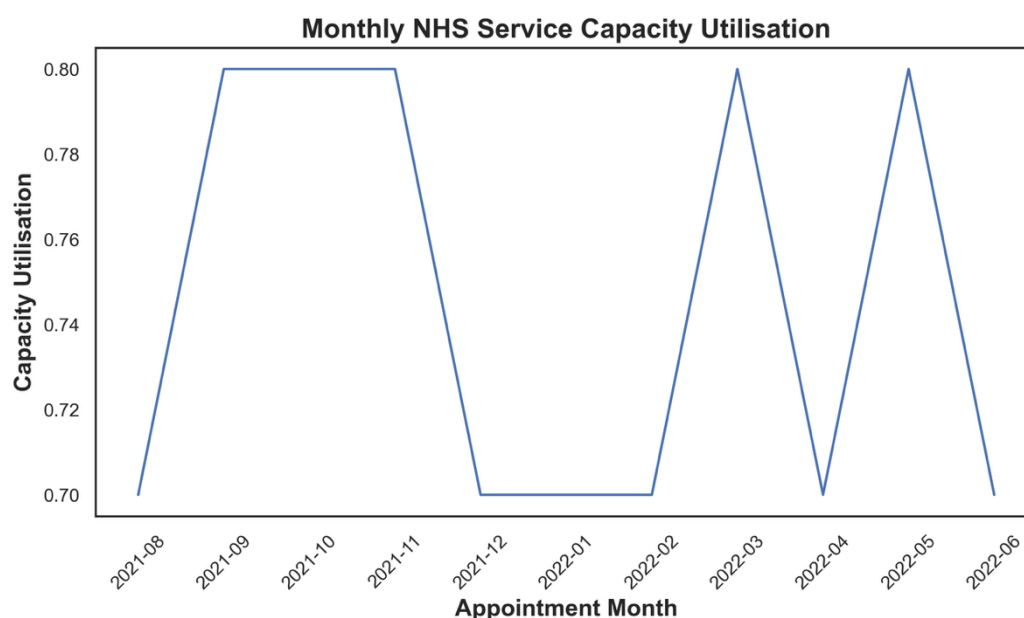


Insights: Offering the patient the option to choose their preferred HCP could enable the reallocation of staff effectively and target the specific needs of the patient. Reviewing and adjusting clinic times to mid-week days, reallocating capacity to suit the demand during the week, would lower wastage of resources. Incorporating automated reminders via SMS, email, or phone calls would reduce the number of missed appointments (DNA). Promoting telephone consultations would also allow for improved service to patients.

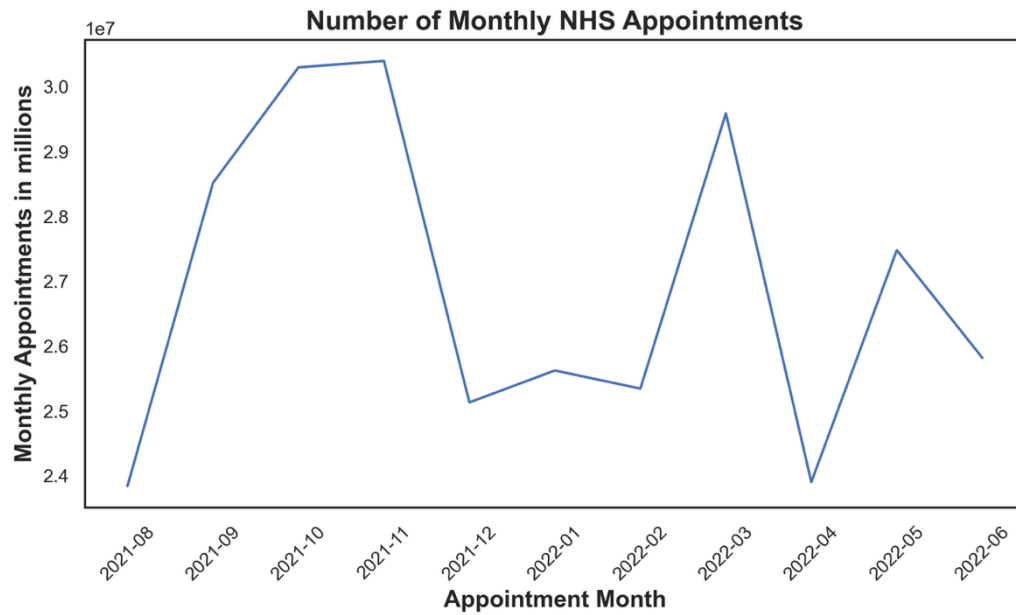
4. Patterns and Predictions



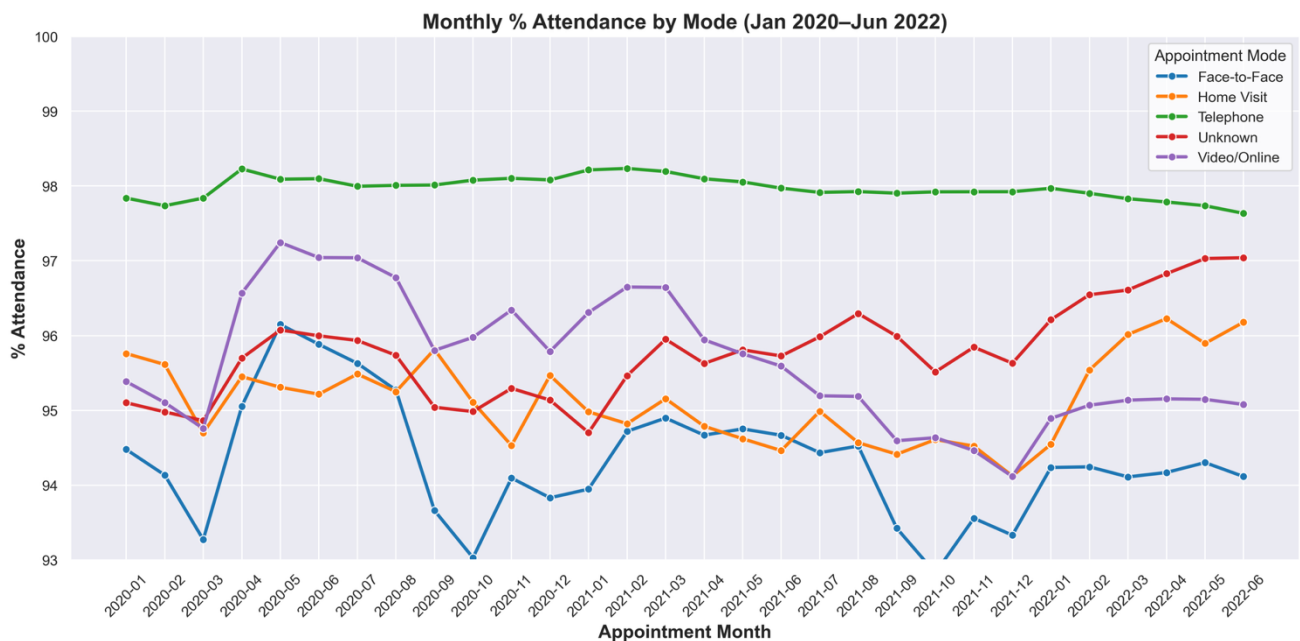
- The capacity utilisation value never exceeds 1 (100%), and the values are stable and relatively high (0.75-0.8 or 75%-80%). This indicates that the NHS is sufficiently staffed and the resources are neither underutilised nor overstretched. Therefore, the NHS does not need to increase its staffing capacity. Instead, it may reallocate the existing resources to aid low-functioning ICB locations like Greater Manchester.

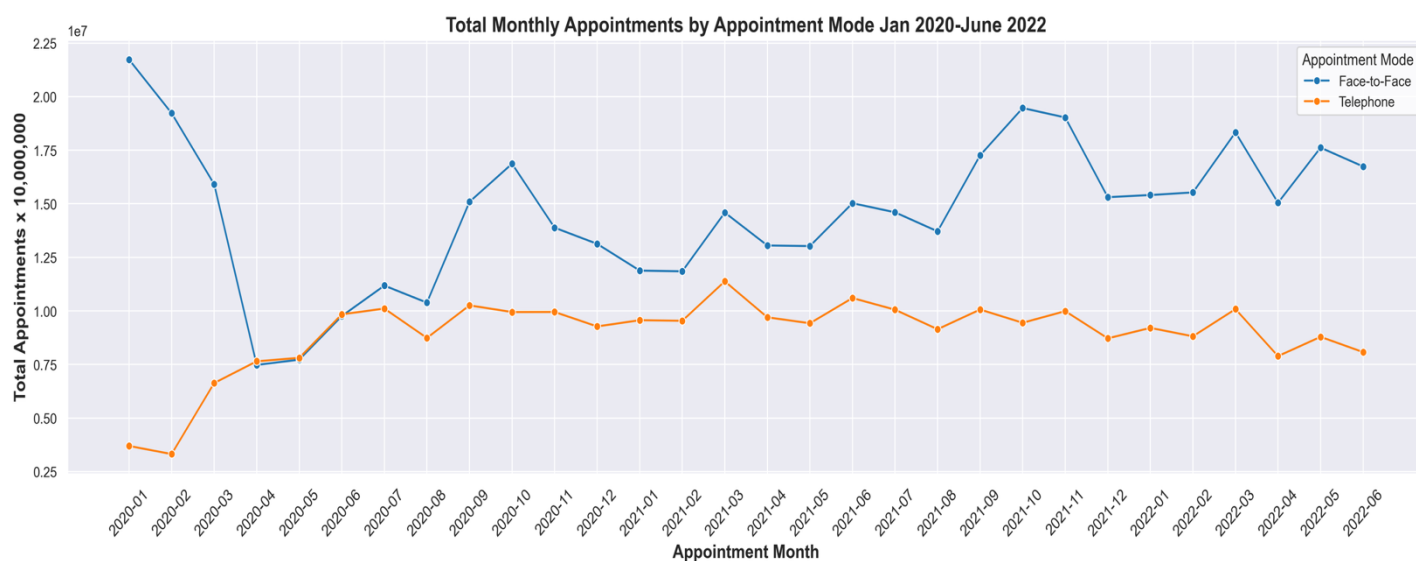


- The NHS was able to handle a large influx of appointments during the COVID-19 pandemic and recover accordingly, as seen in the 'Number of Monthly NHS Appointments' chart below. This indicates that the NHS is adequately staffed and capable of handling high-demand scenarios.

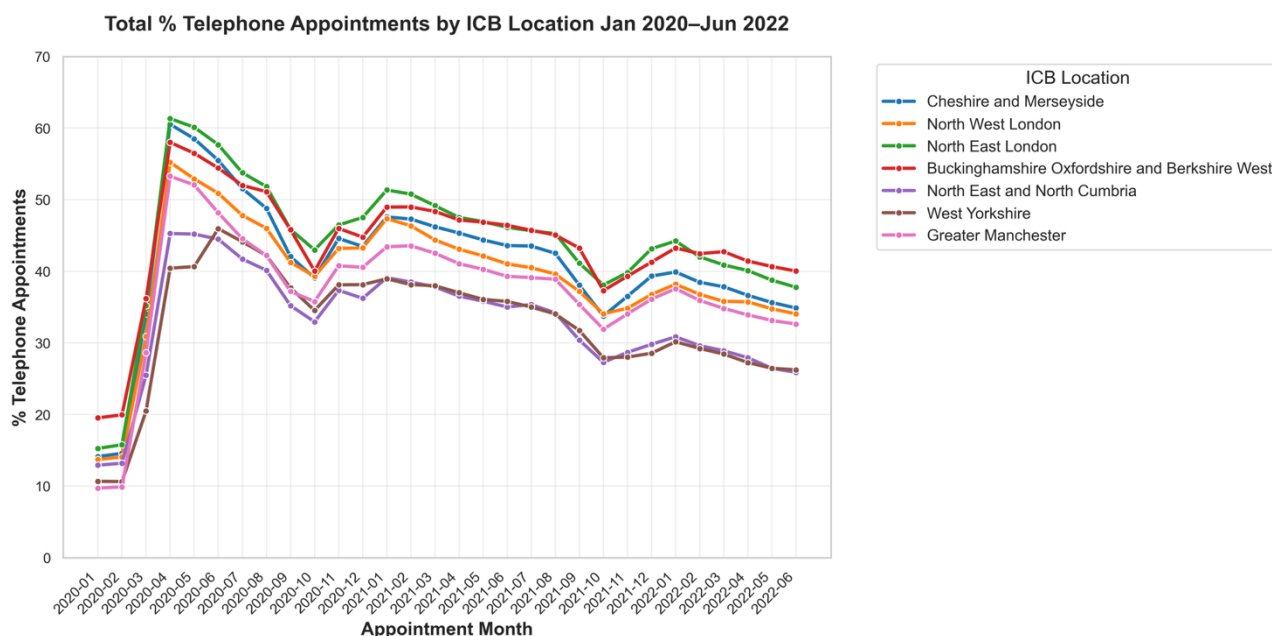


- During the pandemic, direct contact was prohibited and hence, face-to-face appointments had a very high missed rate and saw a downward trend. In response to this, remote/online consultations became a popular mode of appointment. Through effective analysis, it was found that Telephone consultations maintained a steady course of successful attendance throughout the pandemic.

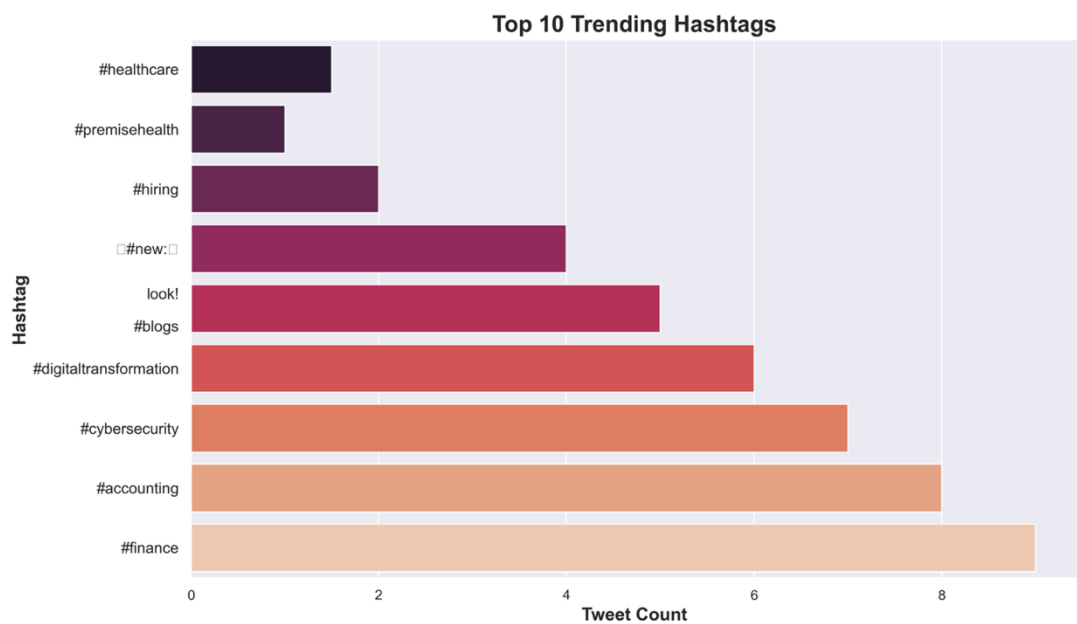




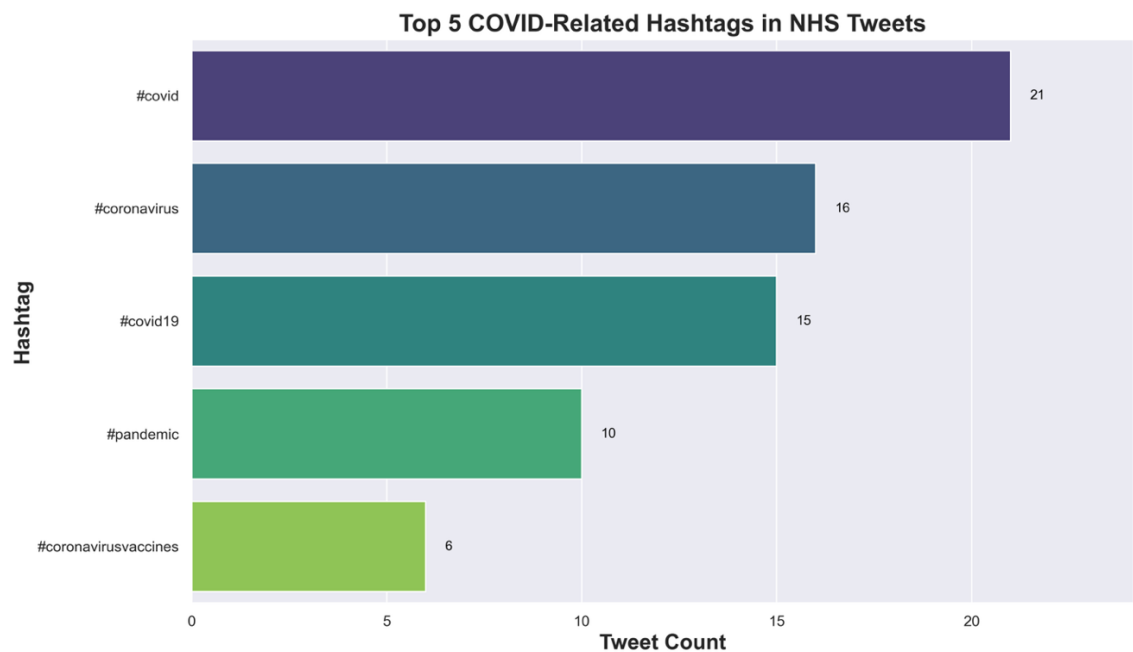
- The percentage of telephone consultations across several NHS regions experienced a rise in popularity at the onset of the COVID-19 pandemic, peaking in early 2020, followed by a gradual decline as in-person care resumed, though rates remained well above pre-pandemic levels. The chart below illustrates the trends. (Appendix 3.9. & 3.10.)



5. Twitter Data Sentiment Analysis



Sentiment analysis of the Twitter data helped discover the top 10 trending hashtags. Further data analysis helped uncover a valuable COVID-19-related theme: The rise of telehealth platforms. The public was very encouraging of the new mode of healthcare post-pandemic as well.



6. Overall Recommendations to the NHS

- Understanding patient behaviour could help identify the underlying causes (socioeconomic factors, communication gaps, access or booking issues, etc) to reduce the number of missed appointments (DNA) in regions with high volumes of missed appointments. This can be implemented through patient surveys, feedback and constant monitoring of appointment data and analysis of the same.
- Analyse high-performing regions (North East and North Cumbria) to learn about the appointment efficiency or adoption of remote modalities to share the best practices NHS-wide.
- Optimise Staffing schedules to align workforce capacity with appointment peaks (weekly/monthly/yearly trends) and anticipated seasonal changes.
- Promote shorter lead times between bookings and appointments, where clinically appropriate and offer easy cancellation/rescheduling options.
- Improve patient communication by deploying an automated reminder system via SMS, emails or calls for regions with high missed appointment rates to avoid wastage of resources and ensure effective utilisation of service capacity.
- Promote a long-term hybrid care model by educating patients and providers about remote consultations (Telephone/Video), targeting settings where in-person visits are in very high demand and regions with high missed appointments.
- Improve the current data quality (inconsistent mapping or unknown data) to gain a better understanding of the data granularity.
- Engage with public feedback regularly, to review patient and public sentiment—such as social media analytics—to identify shifting needs and emerging concerns, and adapt services accordingly.

Appendix: Supplementary Charts and Insights

1. The 5 Whys Framework: (Applied to find the root cause of missed appointments (DNA) among patients)

1. Why are patients missing GP appointments?

Because they do not attend at the scheduled time.

2. Why do they not attend at the scheduled time?

Because they either forget the appointment or face last-minute obstacles.

3. Why do they forget or face obstacles?

Because the reminders are insufficient, or some patients have unpredictable work, family, or transport issues.

4. Why are reminders insufficient, and why do such obstacles disproportionately affect some patients?

Because not all patients receive effective communication (like SMS or phone reminders), and some patients (especially the vulnerable) lack personal or community support/resources to manage or reschedule appointments.

5. Why is communication ineffective, and why is support lacking?

Because current NHS systems may not be fully accessible, personalised, or supportive for all demographic groups, technological barriers, language/cultural differences, or socioeconomic factors may prevent effective engagement with the appointment system.

Conclusion: Improve the quality of resources provided by the NHS to cancel or reschedule appointments, also provide alternate modes of appointments, such that there is ease of access to all patients while also allowing for lower missed appointment rates.

2. Planned Approach for Data Analysis

Approach:

- Import and explore the data provided in the three CSV files.
- Familiarise myself with the variables present in each dataset to answer the questions effectively.
- Identify seasonal trends, daily trends and monthly trends to uncover valuable patterns and insights.
- Carry out sentiment analysis using the Twitter data to understand public sentiment and trends.

- Analyse the resource utilisation using the 1.2 million appointments per day benchmark figure to understand if the current staffing capacity is sufficient.

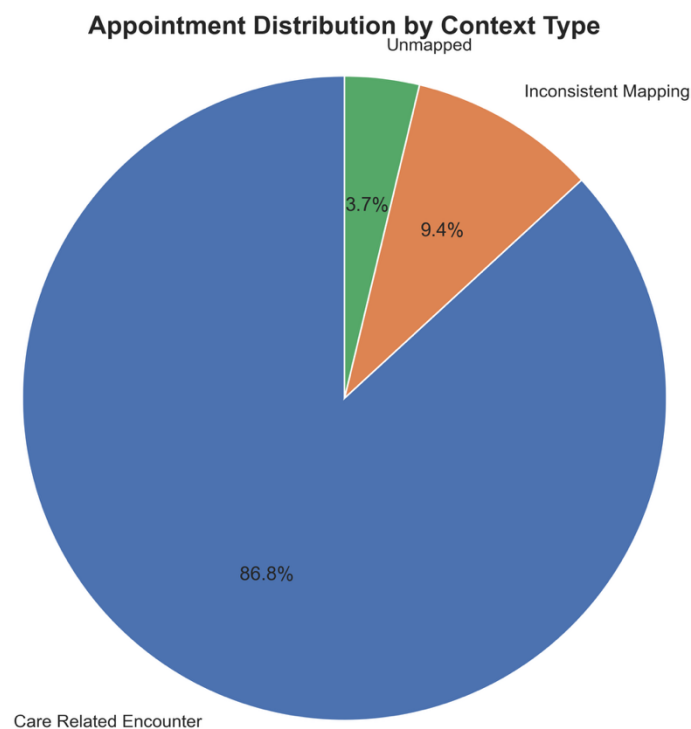
3. Interpretation of Visualisations and Gathering Insights

3.1. Location Statistics of the Data

- NHS England is divided into 7 regions, 42 ICBs (Integrated Care Boards) and 106 sub-ICB locations.
- North East and North Cumbria had the highest number of total appointments throughout the 30 months (43054121).
- Greater Manchester had the highest number of missed appointments (DNA) (1.9 million).

3.2. Service Setting, Context Type and National Category Statistics

- There are 5 Service Settings, 3 Context Types and 18 National Categories
- General Practices were the most popular Service Setting.
- 86.8% of the Context Types were Care Related Encounters.
- General Consultation Routine was the most common National Category



3.3. Overall Monthly Appointment Trends



Fig. 1

- The above line graph allows you to spot the overall increase or decrease in NHS activity, showing the effect of external factors.
- Any sharp dips or spikes are evidence of events impacting the entire system.
- Upward trends indicate rising demand.
- Downward trends or plateaus indicate reduced patient access or seasonal fluctuations.
- November 2021 had the highest number of appointments, and this could be the result of the rise of the Omicron variant during the second wave of the COVID-19 pandemic. Post this uprise, the NHS appointment levels drop (December 2021 – April 2022) and there is a recovery in service volume after the short-term drops, suggesting that the peaks were temporary and the patterns shown in the provided data were influenced by the pandemic.

3.4. Monthly and Seasonal Appointments by Service Setting

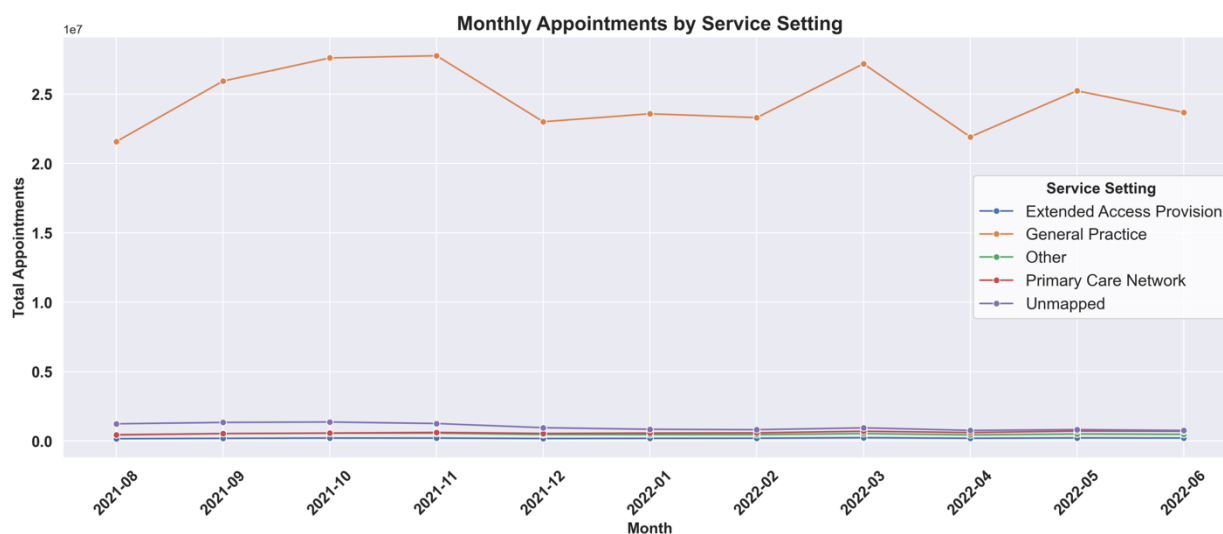


Fig.2

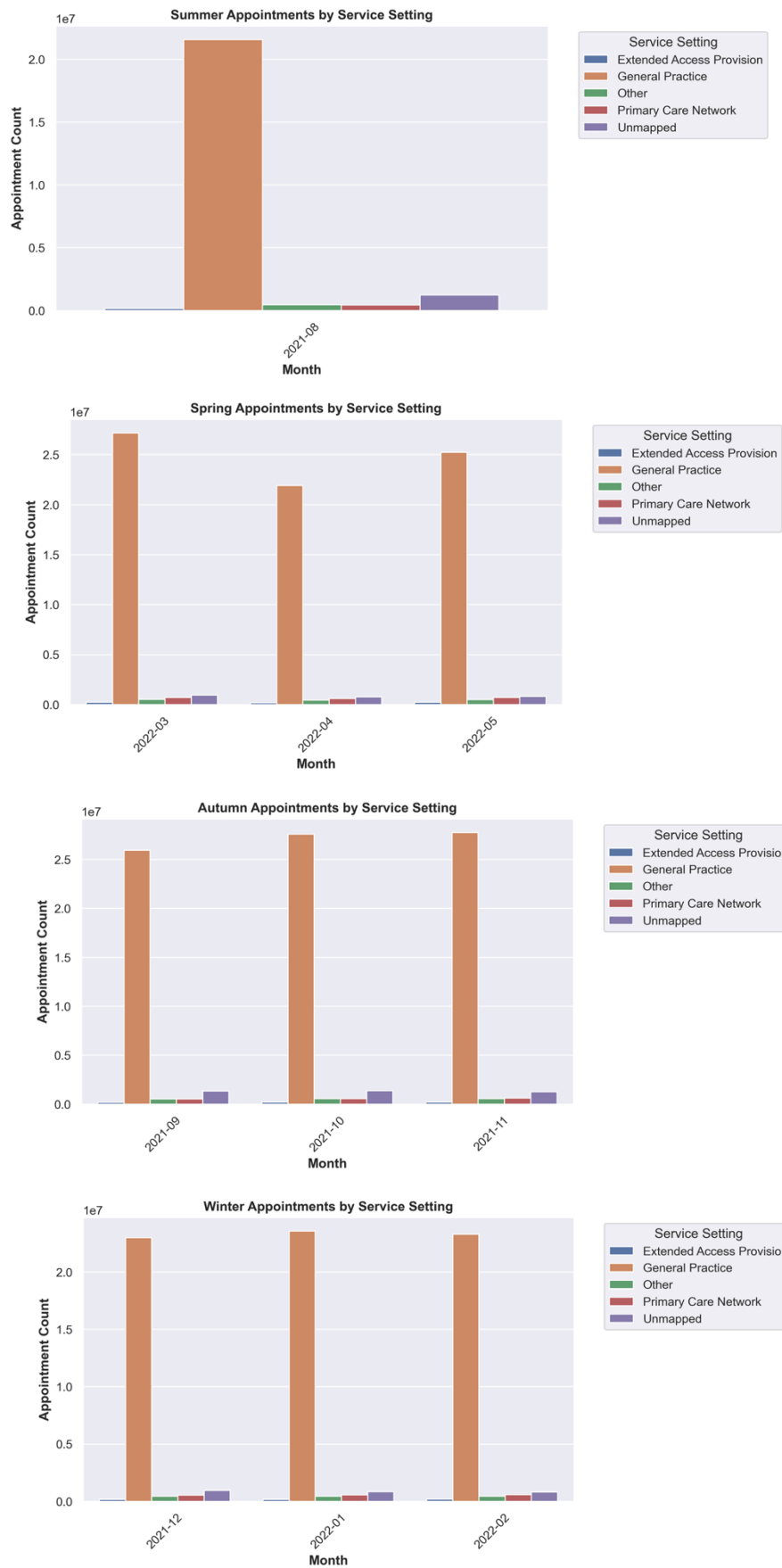


Fig.3

Charts Explained:

Fig.2: The line plot shows the monthly performance of the different service settings over time.

Fig.3: The clustered bar plot shows the seasonal performance across different service settings.

Insights:

- General Practice settings outperform all other service settings, with about 24 million to 30 million appointments per month from August 2021 to June 2022.
- GPs account for about 91.5% of the NHS's total appointments.
- There is a noticeable peak in the number of appointments in November 2021 and a decline during December (Christmas/New Year) and April (Easter), suggesting predictable demand in surges.

3.5. Appointments by Week and Day Type (Weekday vs Weekend)

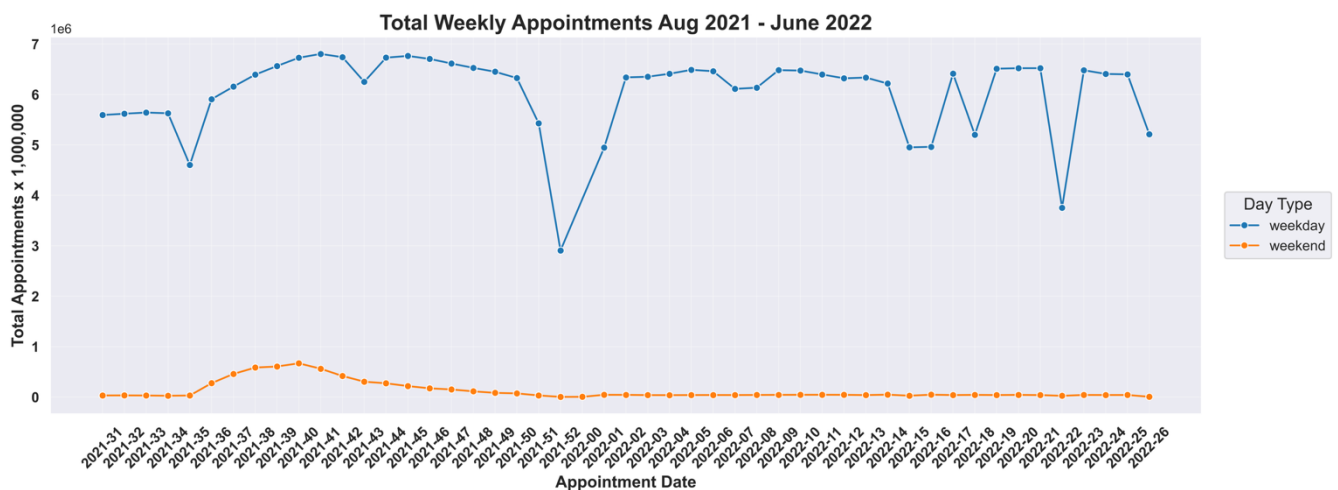


Fig.4

Chart Explained:

Fig.4: The line plot shows the total weekly appointments, with separate lines for weekdays and weekends.

Insights:

- Stable, high appointment volumes on weekdays (5–7 million/week); very low on weekends, indicating both patient and system preference for weekdays.
- The notable declines in total appointments happen at roughly regular intervals, likely aligning with major holidays like Christmas/New Year or statutory breaks.

- No significant growth trend: Both lines remain largely flat, suggesting appointment volumes are not increasing or decreasing systematically over the year

3.6. Appointments by Day of the Week

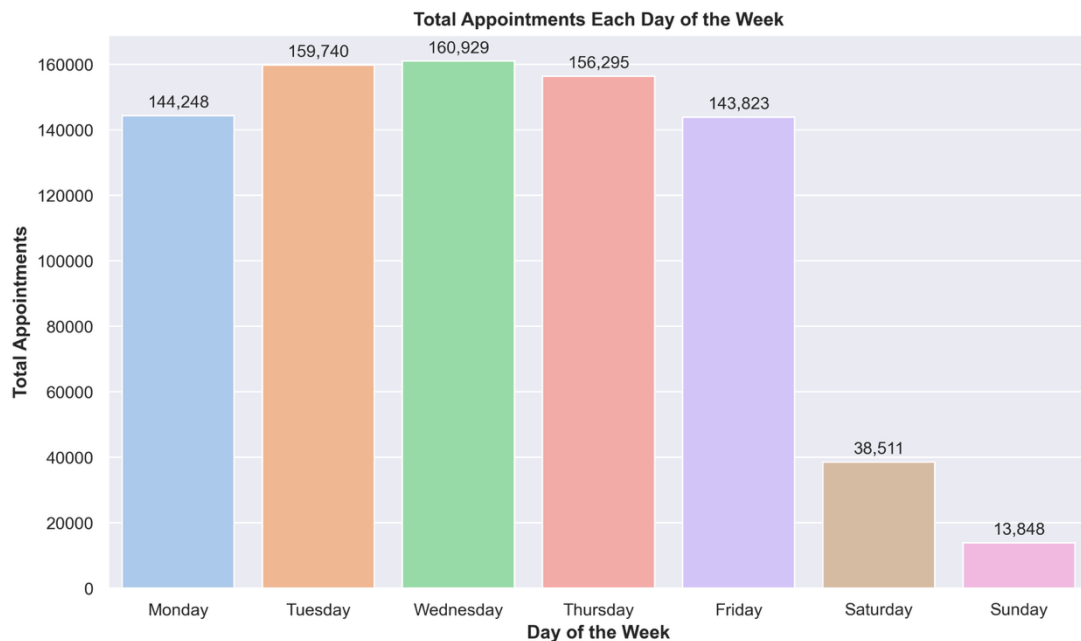


Fig.5

Chart Explained:

The above bar plot shows the number of appointments on each day of the week as per the data contained in the 'national_categories' (nc) DataFrame.

Insights:

- The day of the week chart demonstrates that Monday-Friday have significantly more appointments than weekends, reflecting NHS service models and patient behaviour.
- The highest volume of appointments is mid-week (Tuesday/Wednesday).

3.7. Appointments by National Category

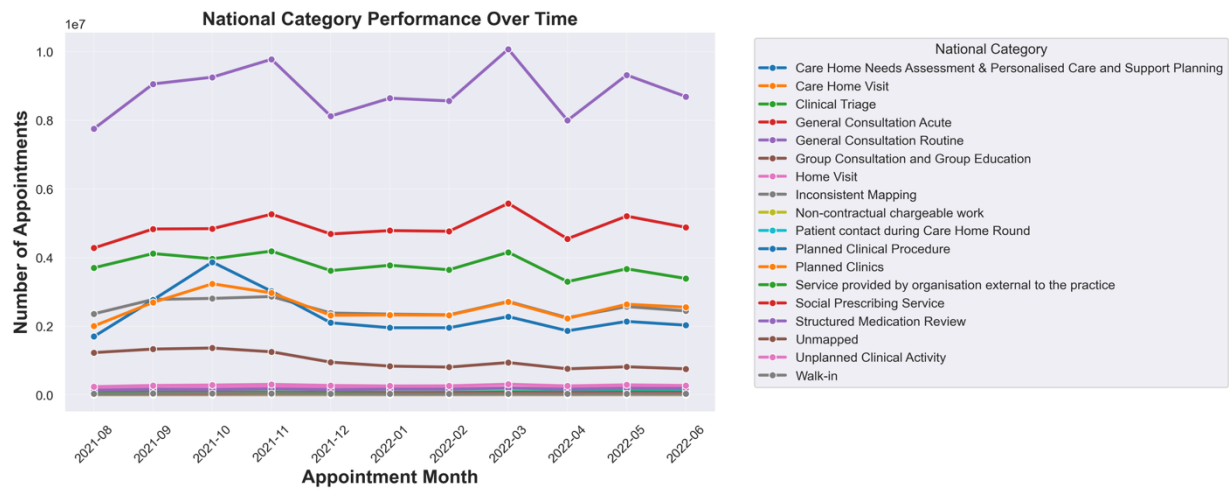


Fig.6

Chart Explained:

The line plot shows the performance of every national category over time.

Insights:

- General Consultation Routine consistently leads, making up a good portion of all appointments.
- The dominance of general consultation routines indicates that most GP appointments are for routine check-ups or ongoing condition management.
- Most national categories show steady trends without major spikes or dips, indicating relatively stable service delivery and demand patterns.
-

3.8. Missed appointments (DNA): Time between the Booking and the Appointment

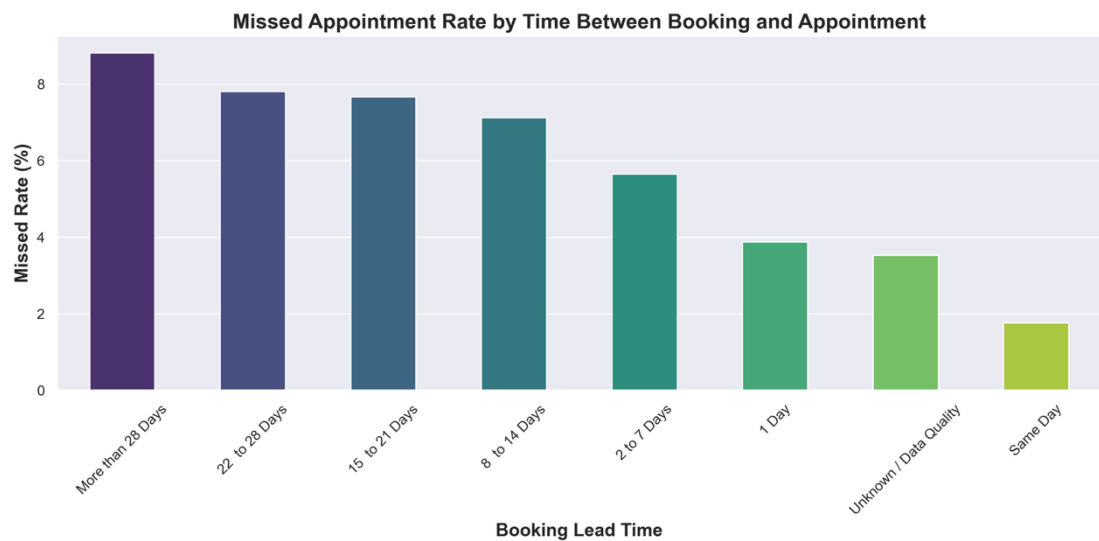


Fig.7

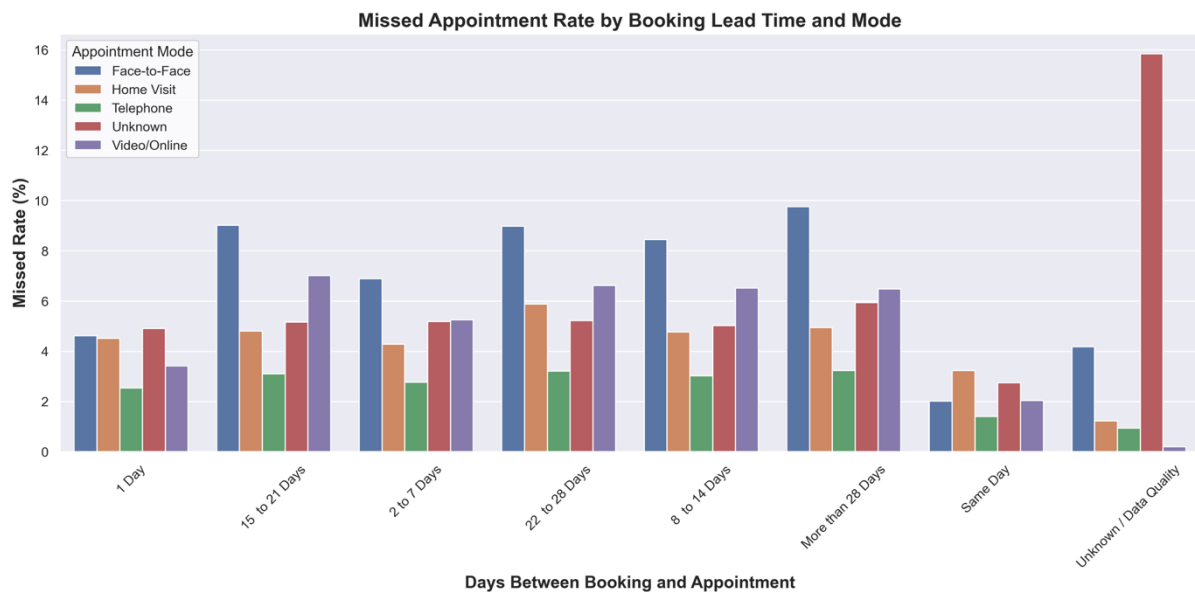


Fig.8

Charts Explained:

Fig.7: The bar plot shows the missed rate of appointments in correlation to the booking lead time of the appointments.

Fig.8: The clustered/grouped bar plot shows the missed rate of appointments in correlation to the booking lead time of appointments across different modes of appointments.

Insights:

- Appointments with a longer wait time between bookings and the appointments saw a higher number of missed appointments (DNA).
- DNAs are lowest for same-day/short-lead appointments; highest when wait times exceed two weeks.
- Face-to-Face appointments saw the highest number of missed appointments (DNA).
- Telephone appointments had the lowest missed rate across all appointment modes.

3.9. Monthly Appointments: Face-to-Face vs. Telephone

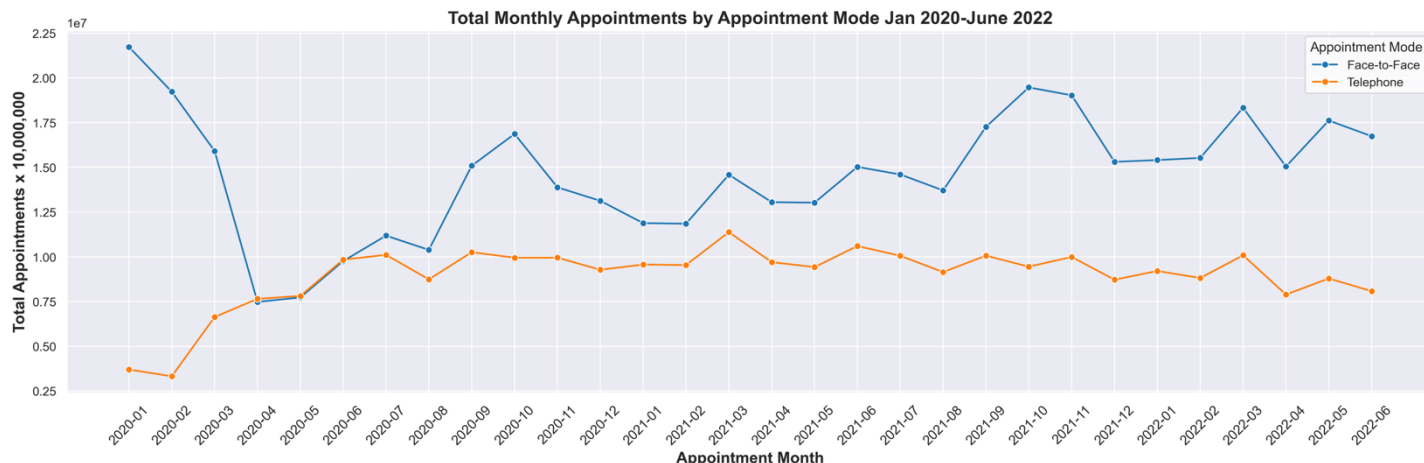


Fig.9

Chart Explained:

The above line plot shows the variation in monthly appointments based on the appointment mode. In this case, Face-to-Face vs. Telephone.

Insights:

- Telephone consultations spiked during the pandemic and have maintained their popularity post-pandemic.
- Telehealth platforms could reduce the number of missed appointments (DNA) and enable access to patients in regions/areas where transport could be an issue.

3.10. Percentage of Telephone Appointments

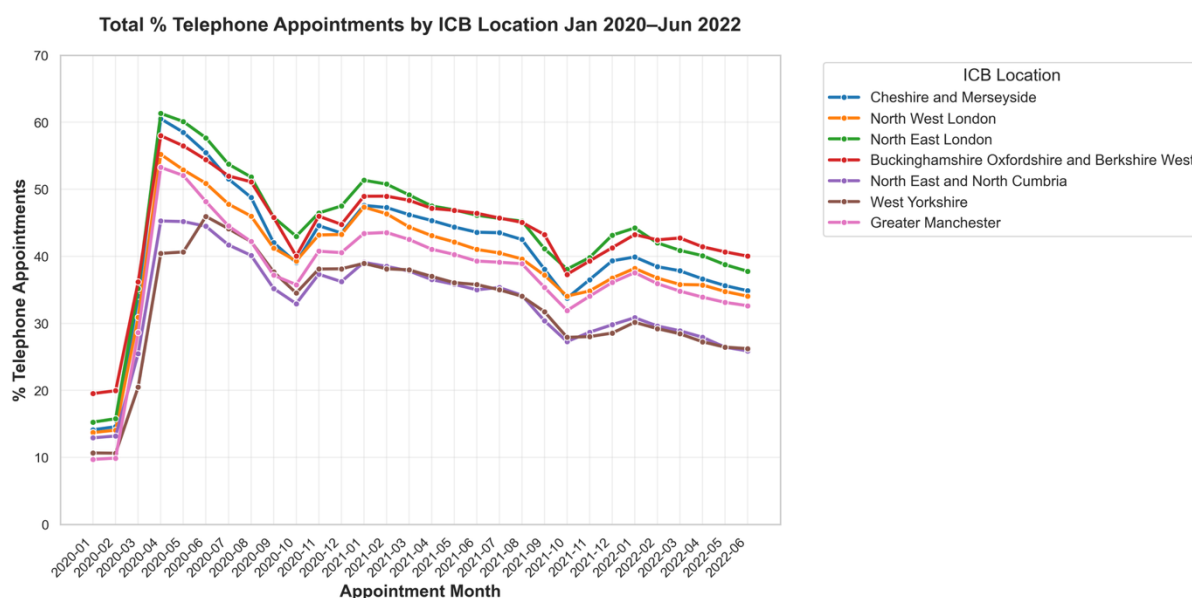


Fig.10

Chart Explained:

The above line plot shows the percentage of telephone appointments by ICB locations from January 2020 to June 2022 (30 months).

Insights:

- A sharp rise in telephone appointments at the onset of the COVID-19 pandemic.
- North East and North Cumbria, Greater Manchester maintained higher percentages compared to some southern or London regions.
- Gradual decline, but not to pre-pandemic levels: After the initial peak, the proportion of telephone appointments generally declined throughout 2021 and 2022, but stabilised at a higher baseline than before the pandemic, indicating a possible lasting shift in practice.

3.11. Monthly Capacity Adequacy Assessment

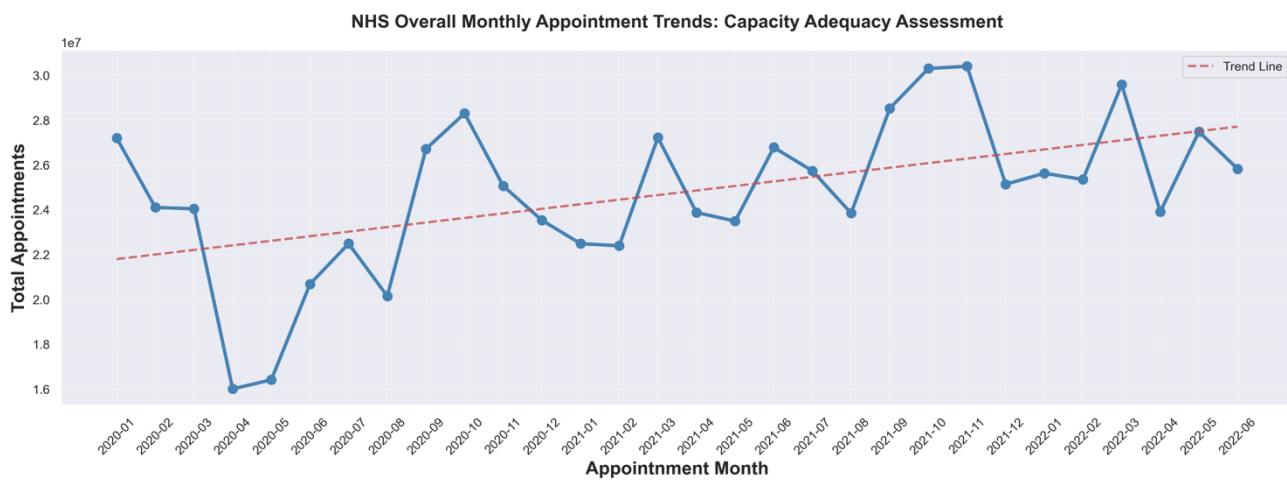


Fig.11

Chart Explained:

The above line plot depicts the total number of appointments from January 2020 to June 2022. The trend line shows positive growth.

Insights:

- The slope of the line is positive, meaning that on average, the number of appointments has been increasing over the 30 months. This indicates a sustained increase in NHS appointments and patient demand.
- Despite major fluctuations caused by COVID-19 waves and holidays, long-term demand rebounded strongly enough to maintain an upward trend.

3.12. NHS Monthly Visits and Service Capacity Utilisation

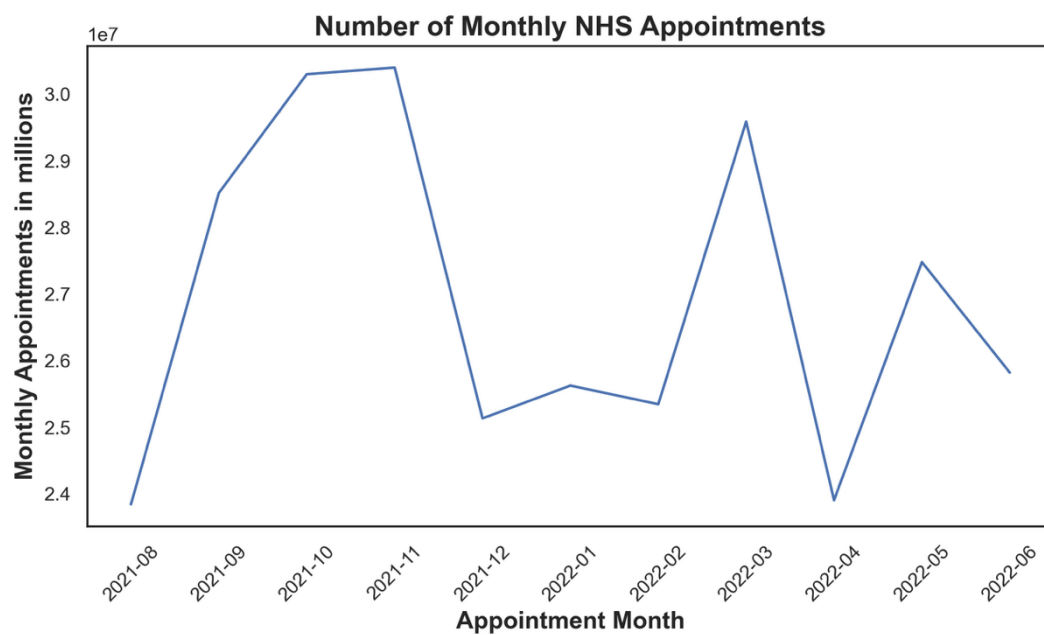


Fig.12

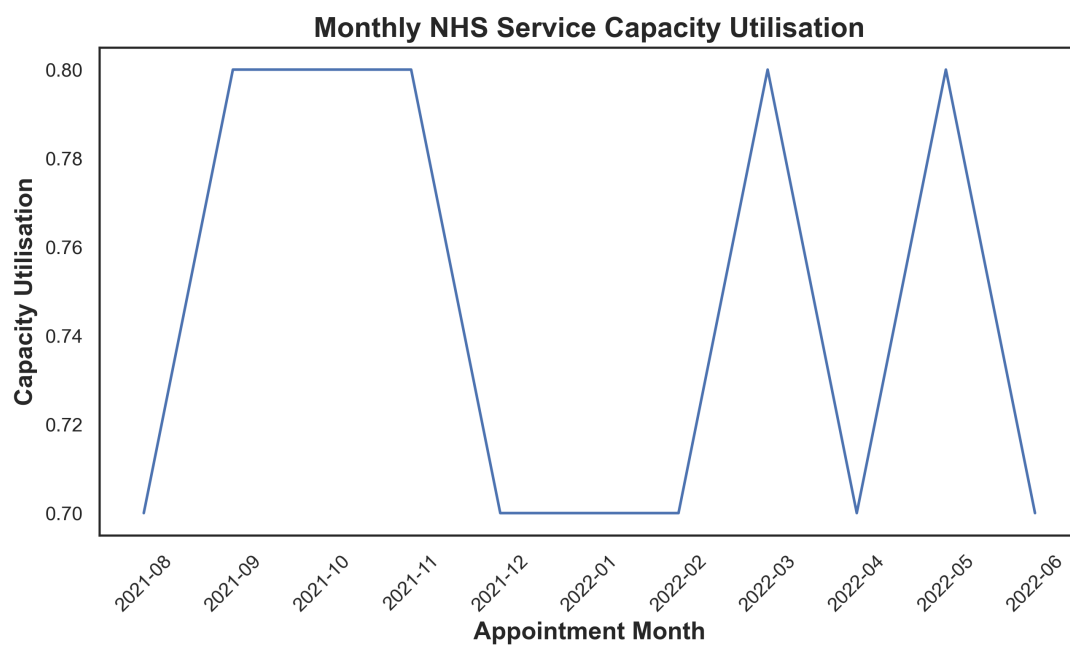


Fig.13

Charts Explained:

Fig.12: The line graph shows the number of monthly NHS appointments in millions.

Fig.13: The line graph shows the NHS Service Capacity Utilisation per month.

Insights:

- The NHS has consistently handled between 24 million and just over 30 million appointments each month from August 2021 to June 2022.
- Noticeable increase in monthly appointments from August to November 2021: could indicate the aftermath of the second wave of the COVID-19 pandemic.
- November 2021 had the highest volume of appointments.
- No evidence of long-term under-performance can be seen.
- The NHS has a benchmark of about 1.2 million appointments per day. In a 31-day month, $1.2 \text{ million} * 31 = 37200000$ appointments. In a 30-day month, $1.2 \text{ million} * 30 = 36000000$ appointments. According to the plot, the NHS is well-staffed to deal with these volumes of appointments.
- The staffing capacity is sufficient, and there is no need to increase the staffing levels.
- From Fig.13, it can be seen that the capacity utilisation never exceeds 1 (100%), suggesting that the NHS's service capacity utilisation values are stable and relatively high (around 0.75-0.8 or 75-80%).
- The NHS is matching its staffing and capacity to patient demand—neither underutilising valuable resources nor overstretched—thus indicating an appropriate and sufficient level of staffing.

4. Twitter Data Sentiment Analysis

4.1 Top 10 Trending Hashtags

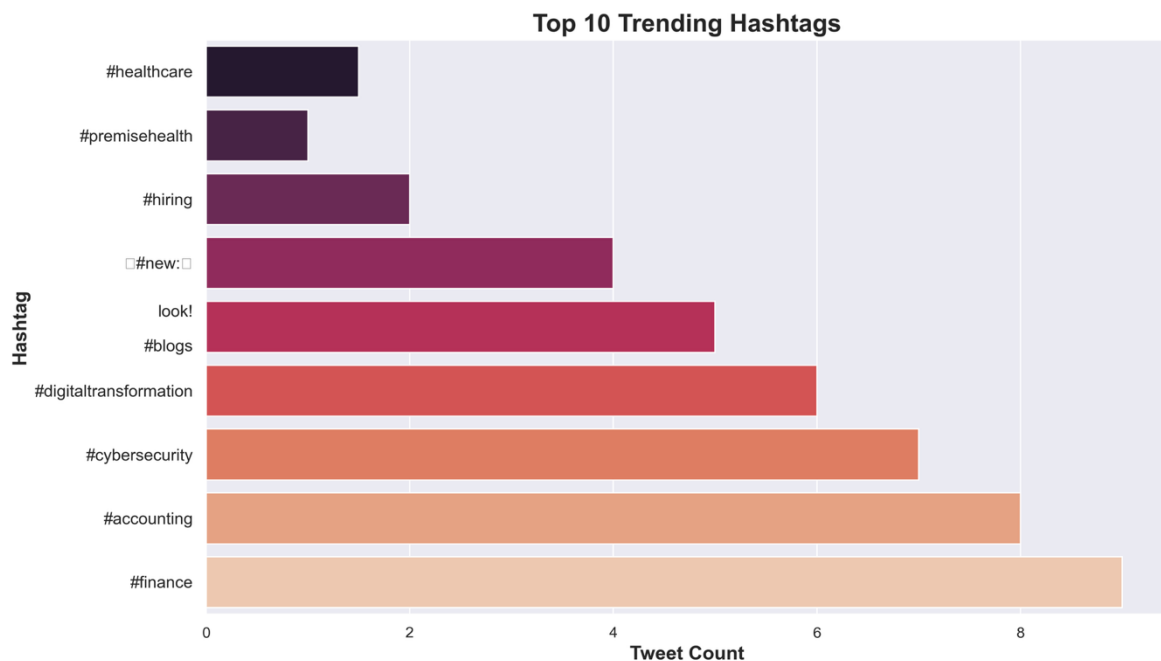


Fig.14

4.2 Top 5 COVID-Related Hashtags in NHS Tweets

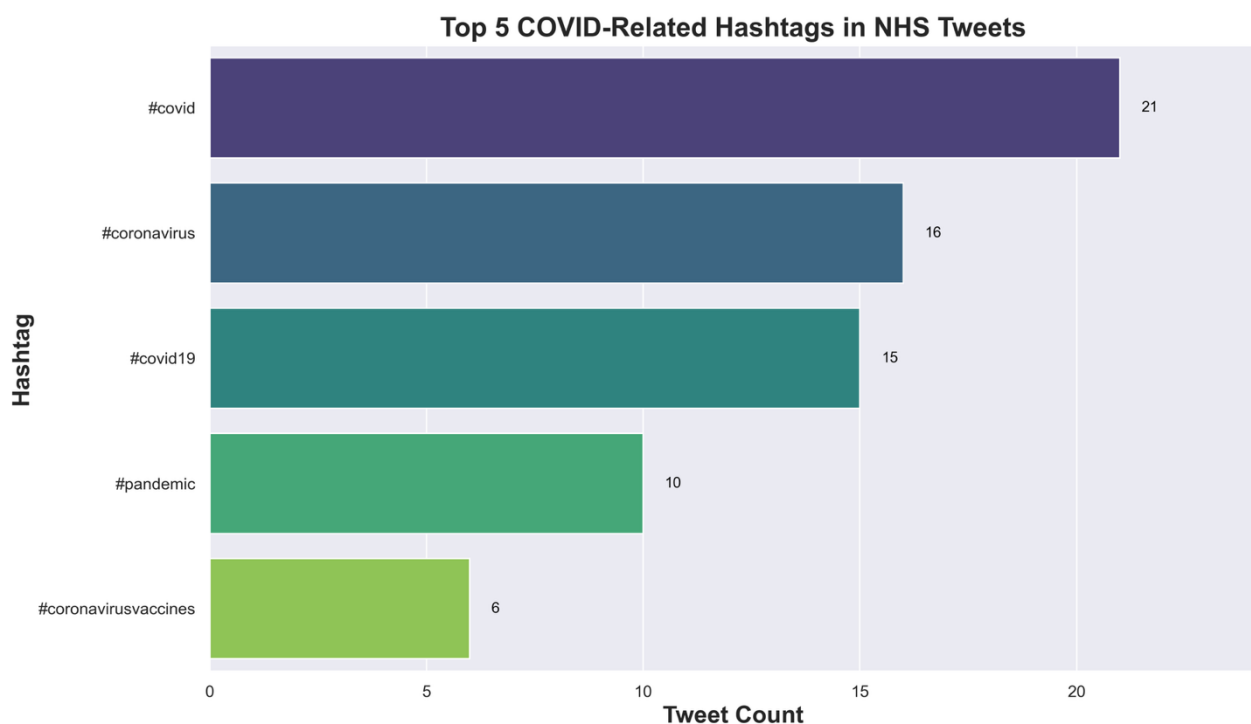


Fig.15

Charts Explained:

Fig.14: The bar chart shows the top 10 trending hashtags found through data scraping of Twitter data.

Fig.15: The bar chart shows the top 5 COVID-related hashtags found through the data scraping of Twitter data.

Insights:

- Through the practice of sentiment analysis, it was found that telehealth platforms saw a rise in popularity during the COVID-19 pandemic. This was discovered through data scraping performed on the Twitter data available.
- There were no tweets or comments regarding staffing issues or incompetence in the NHS system. (no public sentiment around NHS service capacity utilisation)
- There were a few tweets regarding bringing awareness to the importance of cybersecurity and growing technology.